

Introduction to Climate Modeling

Week 5

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Forcing-feedback framework

$$\text{TOA energy balance: } N = Q - \text{OLR}$$

$$N' \approx F + \alpha T'$$

- **Forcing F:** imposed radiation change at zero surface warming.
- **Net climate feedback parameter α :** change in net incoming radiation per kelvin of warming, in $\text{W m}^{-2} \text{K}^{-1}$.
- **T'** is warming from the reference climate.

Positive N' means additional net energy gain.

$\alpha < 0$: restoring response. $\alpha > 0$: amplifying response.

Contributions to the net feedback

IPCC AR6 assessed values, in $\text{W m}^{-2} \text{K}^{-1}$

Feedback contribution	Central estimate	Very likely range
Planck response	−3.22	−3.4 to −3.0
Water vapour + lapse rate	+1.30	+1.1 to +1.5
Surface albedo	+0.35	+0.10 to +0.60
Clouds	+0.42	−0.10 to +0.94
Biogeophysical + non-CO ₂ biogeochemical	−0.01	−0.27 to +0.25
Net feedback	−1.16	−1.81 to −0.51

Positive contributions weaken the net restoring response.
The combined α remains negative.

IPCC AR6 WGI, Table 7.10. Values refer to the assessed net feedback relevant for ECS.

Gregory plots - Lab

Two-box ocean

Fast surface response, slow ocean adjustment

- **Surface reservoir:** the upper ocean and rapidly coupled atmosphere and land, represented by T_s' and heat capacity C_s .
- **Deep reservoir:** the ocean interior, represented by T_d' and a larger heat capacity C_d .
- **Exchange:** as the surface warms, some incoming energy enters the interior and delays surface warming.

Where does the energy go while the planet is still gaining heat?

Two-box ocean: coupled energy budgets

$$C_s \, dT_s'/dt = F + \alpha T_s' - Q'$$

$$C_d \, dT_d'/dt = Q'$$

$$Q' = \gamma(T_s' - T_d')$$

- **Q'** : additional heat transfer from the surface to the deep box, in W m^{-2} .
- **$\gamma \geq 0$** : exchange coefficient, in $\text{W m}^{-2} \text{K}^{-1}$. A larger temperature-anomaly difference drives more transfer.
- **C_s and C_d** : heat capacities in $\text{J m}^{-2} \text{K}^{-1}$, both per total Earth area.

The surface loses exactly the energy that the deep box gains.

Two-box ocean: equilibrium and adjustment

$$N' = C_s dT_s'/dt + C_d dT_d'/dt$$

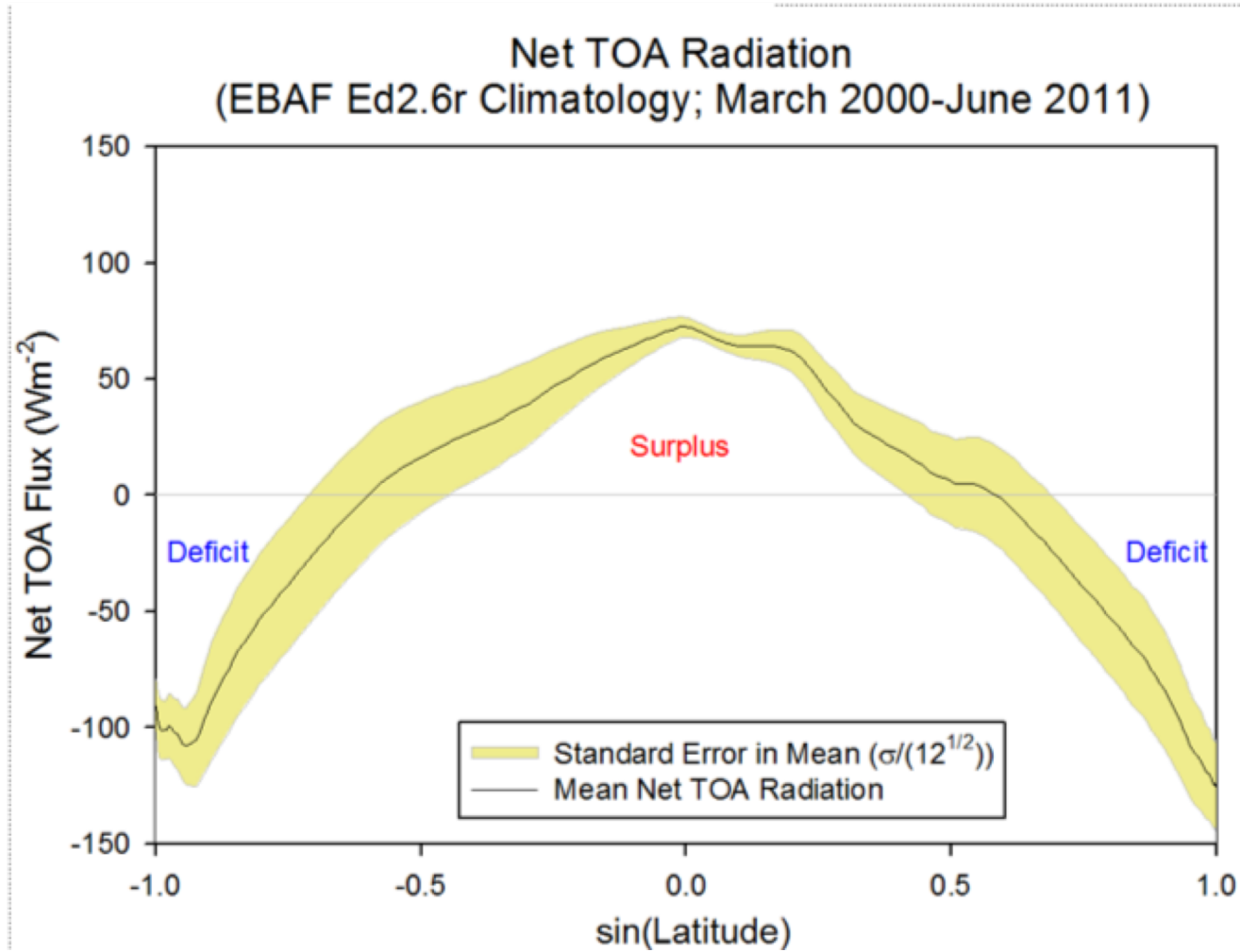
$$\text{At equilibrium: } T_s' = T_d' = -F/\alpha \quad (\gamma > 0)$$

- **Conservation:** internal exchange cancels from the total energy budget.
- **Equilibrium:** vanishing deep storage requires $Q' = 0$. Both warming anomalies eventually match.
- **Adjustment:** exchange and deep heat capacity change the path and pace of warming, while F and α set the endpoint.

Ocean notebook: compare one and two reservoirs, then vary γ and C_d .

Two-box - Lab

Observed regional radiation budget



Net radiation = absorbed sunlight – outgoing infrared radiation

The axis runs from the South Pole (–1), through the equator (0), to the North Pole (+1).

What keeps the tropics from continually warming and the higher latitudes from continually cooling?

Source: NASA CERES Science. March 2000–June 2011 climatology.

Tropical and polar boxes

Unequal sunlight, connected regional climates

- **Tropical box:** the belt from 30°S to 30°N, with stronger absorbed sunlight.
- **Polar box:** the remaining higher latitudes, with the same total area and weaker sunlight.
- **Heat transport:** atmosphere and ocean move energy between the regions, allowing a radiative surplus in one and a deficit in the other.

How can the poles stay at a steady temperature while losing energy by radiation?

Regional radiation and heat transport

$$\begin{aligned} N_i &= S_i - R(T_i) & S_i &= (1 - \mathcal{A}_i) I_i \\ R(T_i) &= R_0 + B(T_i - T_0) & Q &= \kappa(T_t - T_p) \end{aligned}$$

- **Regional radiation:** I_i is incoming sunlight and $\mathcal{A}_i$ is albedo. S_i and R are fluxes in W m^{-2} .
- **Outgoing radiation:** $R_0 = 210 \text{ W m}^{-2}$ at 0°C . $B = 2 \text{ W m}^{-2} \text{ K}^{-1}$ sets the increase in emission per degree.
- **Transport Q:** poleward flux in W m^{-2} . The coefficient $\kappa \geq 0$ has units $\text{W m}^{-2} \text{ K}^{-1}$.

With fixed sunlight, the net feedback is $\alpha = -B < 0$.

Tropical and polar energy budgets

$$C_t \, dT_t/dt = S_t - R(T_t) - Q$$
$$C_p \, dT_p/dt = S_p - R(T_p) + Q$$

- **Equal areas:** each region covers half of Earth.
- **Heat capacities C_t and C_p :** energy per kelvin per unit area of each region.
- **Exchange:** the same flux Q leaves the tropics and enters the higher latitudes. It cancels when we add the budgets.

Global means are simple averages of the two equal-area regions.

Steady temperatures with continuing heat transport

$$N_t = Q > 0 \quad N_p = -Q < 0$$
$$(N_t + N_p) / 2 = 0$$

- **Regional equilibrium:** exported heat balances the tropical radiative surplus.
Imported heat balances the polar deficit.
- **Global equilibrium:** internal transport cancels when we average the two equal-area budgets.
- **Transport notebook:** vary κ and compare regional warming and cooling.

Steady temperature requires zero total storage in each region.

Regional climate - Lab